Ruilin (Peter) Xie

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Education

University of California, Irvine Irvine, CA
Master of Software Engineering Sep 2024 – Dec 2025
Coursework: Data Structures/Algorithms, Distributed Architecture, Software Testing & Debugging, Reverse Engineering, Concurrent/Network Programming

University of California, Irvine
Bachelor of Arts, Business Economics Jun 2018 – Aug 2022

Skills

Languages: C# · C++ · Python · Blueprint (UE5) · Java · TypeScript
Engines & Tools: Unity 2022 · Unreal Engine 5 · UMG · Git · Diversion
Disciplines: Gameplay Systems · Runtime Architecture · Data-Driven Design · Technical Design · Balance Tooling

Projects

CrossBolt — Rhythm Game Prototype Unity 2022, C#, | Team (4) | 2025
`github.com/MisakaRinOwO/Crossbolt-Code-Samples`
- Designed data-driven note schema, chart addressing, and custom .txt grammar consumed by runtime loader and editor tooling
- Implemented chart pipeline: NoteLevelConverter → typed timestamp queue → ChartSpawner with type-aware prefab selection and O(1) hit lookup
- Built Unity Editor beat-map editor; optimized judgement branch from O(N) active-list scan to O(P) pending-window checks + O(1) hold lookups

Grid Systems Experiment — UE5 C++ Gameplay Systems UE5, C++, Blueprint | Solo | 2026
`github.com/MisakaRinOwO/UE5-Grid-Experiment`
- Built reusable `AGridManager` C++ core: `TArray`-based cells, world/grid conversion, cost-aware A*, movement-budget expansion, occupancy, and debug visualization
- Separated native grid/pathfinding logic from scenario rules; Tactical, Weighted Terrain, and Tower Defense levels reuse the same core through Blueprint PlayerControllers
- Implemented gameplay-facing validation flows: hover path-cost preview, Data Asset-driven unit movement/ability ranges, and obstacle rollback when Tower Defense routes are blocked

Void Architect — Modular Ship Combat Prototype UE5, Blueprint, Python | Solo | 2026
- Implemented tick-driven combat simulation: tracking vs. agility hit probability, weapon cooldowns, shield→HP overflow, and replay event recording
- Built equilateral-triangle placement grid with orientation-aware adjacency rules; data-driven module architecture via UE5 Data Assets with CPU/power budget constraints
- Python aggregation scripts identified Sniper shield outlier; nerfed values and verified win-rate spread across full build matrix with replay UI inspection

Project Summon — Action RPG Capstone UE5, Blueprint, C++ | Team (5) | 2025
- Shipped complete gameplay loop (hub → room combat → loot → return) on fixed 10-week schedule; owned room progression, combat, inventory, and persistence systems
- Built BP_RoomManager as room authority: overlap-scoped enemy tracking, O(1) clear check, and enemy/chest/portal spawning on clear trigger
- Integrated teammate C++ enemy code by exposing HP/TakeDamage (`UPROPERTY/UFUNCTION`) and implemented GameInstance-based cross-level player data sync

Godot Engine — Open Source C++ Contributions C++, Godot 4 | 2025
`github.com/godotengine/godot/pulls?q=is%3Apr+author%3AMisakaRinOwO`
- PR #107101 fixed CSGShapes3D hover feedback; PR #107296 added 3D Ruler helper lines/labels; both passed CI and maintainer review